Ex.no:6b	DATA VISUALIZATION USING POWER BI

Aim:

To Load the data into PowerBI and transform the data and visualize it to get insights

Algorithm:

Algorithm: Sales Dashboard Calculation

Step 1: Input Data Loading

Read the dataset from diabetes_prediction_dataset.csv containing columns gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level, and diabetes.

Dataset contains 100,000 patient records.

The diabetes column represents the diabetes status:

0 → No Diabetes

1 → Diabetes

The dataset contains 8,500 diabetes cases and 91,500 non-diabetes cases.

Step 2: Smoking History Filter Check

Check slicer selection for Smoking History.

Available smoking-history categories:

current

ever

former

never

No Info

not current

Default selected categories:

current

ever

former

never

No Info

not current

Retain all patient records matching the selected smoking-history categories in Filtered_Data.

Step 3: KPI Total Diabetes Cases Calculation

Count all records where diabetes = 1 from Filtered_Data.

Total Diabetes Cases:

8,500

The KPI card displays the value as:

8.5K

Total patients in the original dataset:

100,000

Diabetes percentage:

8.50%

Step 4: Bar Chart Calculation

Number of Diabetes Cases by Smoking History

Group Filtered_Data by smoking_history.

Count the records where diabetes = 1 for each smoking-history category.

The bar chart displays the number of diabetes cases for:

never

No Info

current

former

not current

ever

Sort the smoking-history categories in descending order according to the number of diabetes cases.

The resulting bar chart represents the distribution of diabetes cases across different smoking-history categories.

Step 5: Donut Chart Calculation

Number of Diabetes Cases by Gender

Group Filtered_Data by gender.

Count the records where diabetes = 1 for each gender category.

Calculate the percentage share of each gender relative to the total diabetes cases.

Formula:

Gender Share = (Diabetes Cases for Gender / Total Diabetes Cases) × 100

The donut chart displays diabetes cases for:

Female

Male

Other

The center of the donut chart displays:

8.5K

The total represents the overall number of diabetes-positive patients.

The percentage labels represent each gender's contribution to the total diabetes cases.

Step 6: Dashboard Render

Render the KPI card with the formatted total diabetes cases:

8.5K

Render a vertical bar chart showing:

Number of Diabetes Cases by Smoking History

Render a donut chart showing:

Number of Diabetes Cases by Gender

Render a Smoking History slicer containing:

current

ever

former

never

No Info

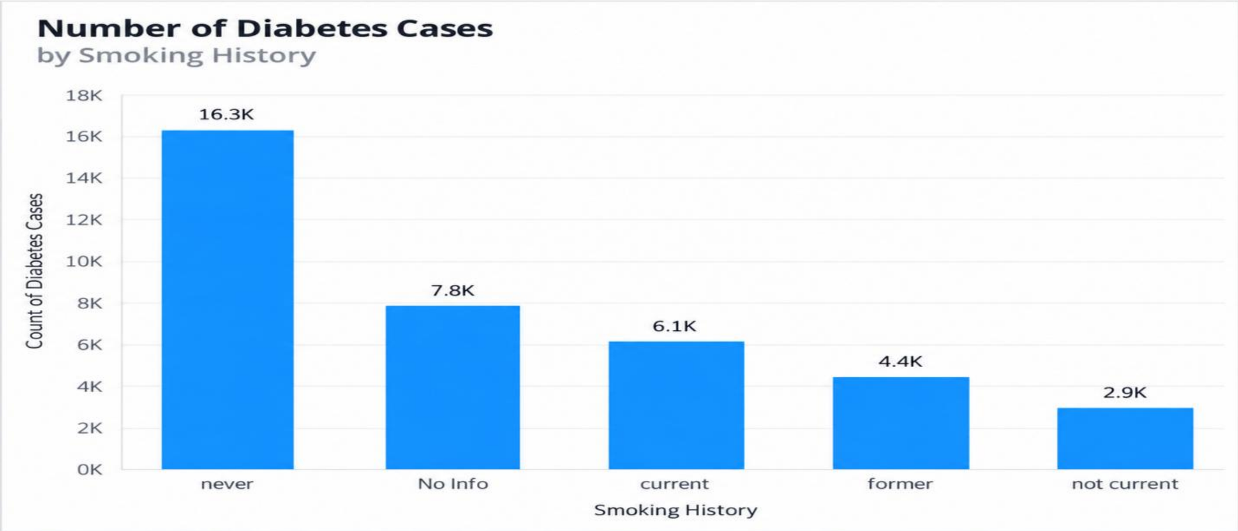
not current

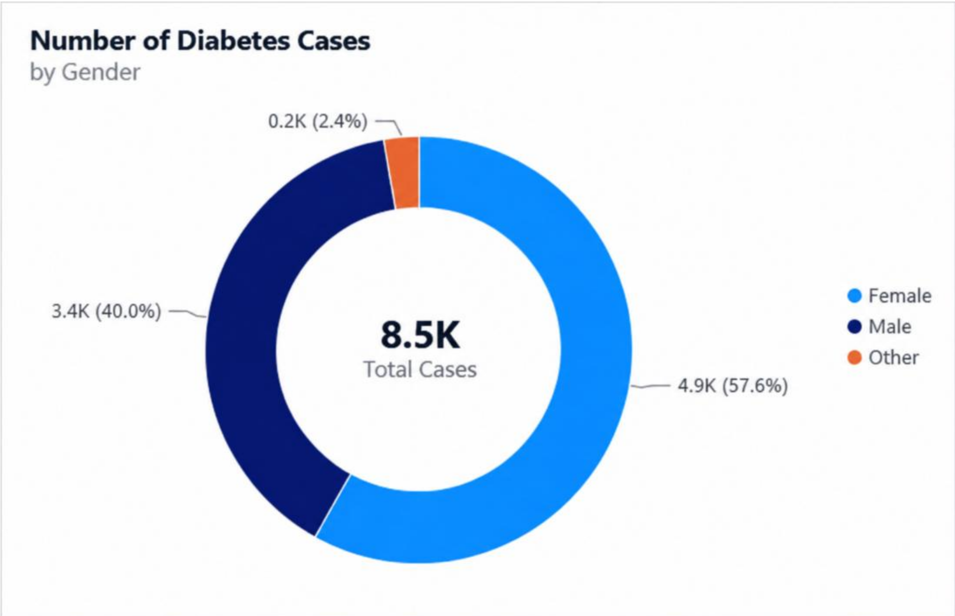
On slicer selection, filter the patient records according to the selected smoking-history categories.

Recalculate the KPI value, bar chart, and donut chart based on the filtered data.

The dashboard therefore provides an interactive analysis of diabetes cases according to smoking history and gender.

Output:


Number of Diabetes Cases
by Smoking History
18K
16K
14K
12K
10K
8K
6K
4K
2K
0K
Count of Diabetes Cases
16.3K
7.8K
6.1K
4.4K
2.9K
never
No Info
current
former
not current
Smoking History


Number of Diabetes Cases
by Gender
0.2K (2.4%)
3.4K (40.0%)
8.5K
Total Cases
4.9K (57.6%)
Female
Male
Other


Total Diabetes Cases
8.5K
Smoking History
current
former
never
No Info
not current

Result:

Result: